第二十四讲

卡氏定理

作业：13-7，8，9，12，13.

第十三章　能量法

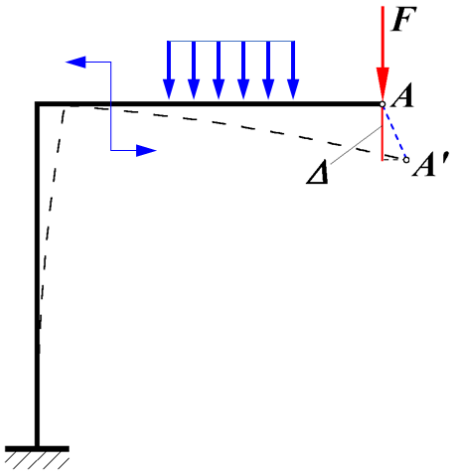
13.1 外力功与应变能

13.2 功的互等定理

13.3 卡氏定理

13.4 虚功原理与单位载荷法

上讲回顾



相应位移

- 载荷作用点处，沿载荷方向的位移；
- 由所有载荷共同引起；
- 广义载荷与广义位移。

克拉珀龙(Clapeyron)定理

$$W=\sum_{i=1}^{n}\frac{F_i\Delta_i}{2}$$

- 线弹性体的外力功与加载次序无关；
- 外力功不可叠加。

弹性体的功能原理

- 应变能是载荷的函数。

$$\boxed{V_\varepsilon=W}$$

$$V_\varepsilon=\int_l\frac{F_N^2(x)\mathrm{d}x}{2EA}+\int_l\frac{T^2(x)\mathrm{d}x}{2GI_t}+\int_l\frac{M_y^2(x)\mathrm{d}x}{2EI_y}+\int_l\frac{M_z^2(x)\mathrm{d}x}{2EI_z}$$

功的互等定理

$$F_1\Delta_{12}=F_2\Delta_{21}$$

$$\sum_{i=1}^{n}F_i\Delta_{iP}=\sum_{j=1}^{m}P_j\Delta_{jF}$$

- 适用于线弹性体；
- 存在两种受力状态。

第二十四讲

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第十三章　能量法

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第十三章 能量法

13.3 卡氏定理

1、卡氏定理的直接推导

2、卡氏定理应用

3、余功与余能

4、克罗第-恩格塞定理

5、例题

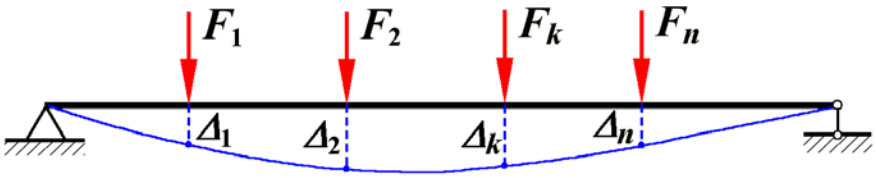
单辉祖编著，材料力学（II）（第1版），北京：高等教育出版社，1999

1、卡氏定理的直接推导

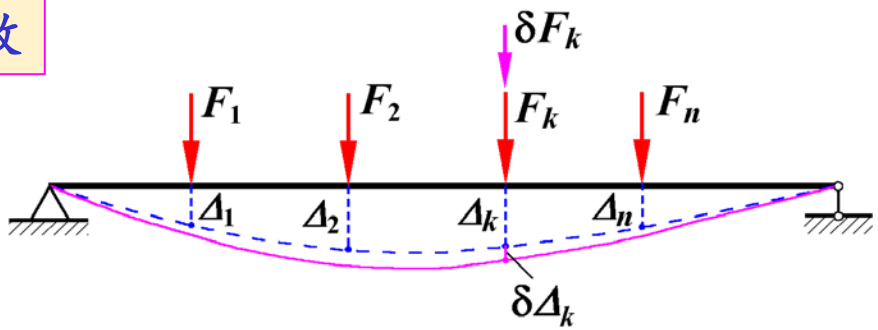
线弹性体受n个载荷

$$V_0 = W_0 = \sum_{i=1}^{n} \frac{F_i \Delta_i}{2}$$

是载荷的函数



给载荷增量 δF_k



考虑两个加载过程

1) 先F_i，后δF_k，应变能

$$V_1 = V_0 + \frac{\partial V_\varepsilon}{\partial F_k} \delta F_k$$

2) 先δF_k，后F_i，应变能

$$V_1' = W_1' = \frac{\delta F_k \delta \Delta_k}{2} + W_0 + \Delta_k \delta F_k$$

线弹性体外力功（应变能）与加载次序无关

$$\Delta_k = \frac{\partial V_\varepsilon}{\partial F_k}$$

1、卡氏定理的直接推导

$$\Delta_k = \frac{\partial V_\varepsilon}{\partial F_k}$$

线性弹性体的应变能，对载荷 F_k 的偏导数，等于该载荷的相应位移 Δ_k

——卡氏第二定理

$$V_\varepsilon = \int_l \frac{F_N^2(x)\mathrm{d}x}{2EA} + \int_l \frac{T^2(x)\mathrm{d}x}{2GI_t} + \int_l \frac{M_y^2(x)\mathrm{d}x}{2EI_y} + \int_l \frac{M_z^2(x)\mathrm{d}x}{2EI_z}$$

$$\Delta_k = \int_l \frac{F_N(x)}{EA}\frac{\partial F_N(x)}{\partial F_k}\mathrm{d}x + \int_l \frac{T(x)}{GI_t}\frac{\partial T(x)}{\partial F_k}\mathrm{d}x + \int_l \frac{M_y(x)}{EI_y}\frac{\partial M_y}{\partial F_k}\mathrm{d}x + \int_l \frac{M_z(x)}{EI_z}\frac{\partial M_z}{\partial F_k}\mathrm{d}x$$

A. Castigliano (1847-1884)，意大利工程师。**1870**年入都灵工业学院，**1873**年提出工程师学位论文。

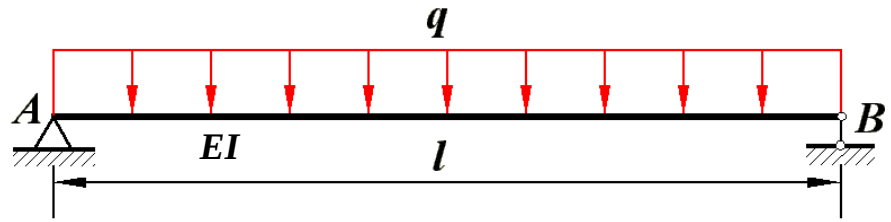
- 卡氏第一定理（弹性体）

$$F_k = \frac{\partial V_\varepsilon}{\partial \Delta_k}$$

2、卡氏定理应用

1）附加载荷

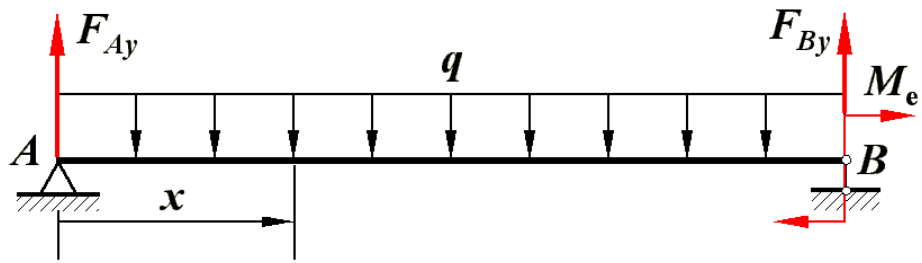
例13-5 图示梁，EI为常数，求θ_B。



解： 施加力矩M_e

$$\theta_B(q,M_e)=\frac{\partial V_\varepsilon}{\partial M_e}$$

$$\theta_B(q)=\theta_B(q,M_e)\big|_{M_e=0}$$



附加载荷

$$F_{Ay}=\frac{ql}{2}-\frac{M_e}{l}$$

$$M(x)=\frac{qlx}{2}-\frac{M_e x}{l}-\frac{qx^2}{2} \qquad \frac{\partial M}{\partial M_e}=-\frac{x}{l}$$

$$\theta_B(q)=\frac{1}{EI}\int_0^l\left(\frac{qlx}{2}-\frac{qx^2}{2}\right)\left(-\frac{x}{l}\right)\mathrm{d}x=-\frac{ql^3}{24EI} \quad (\circlearrowleft)$$

2、卡氏定理应用

2）区分载荷

例13-6 图示梁，EI为常数，求：w_A

解： $M(x)=Fx+Fl$

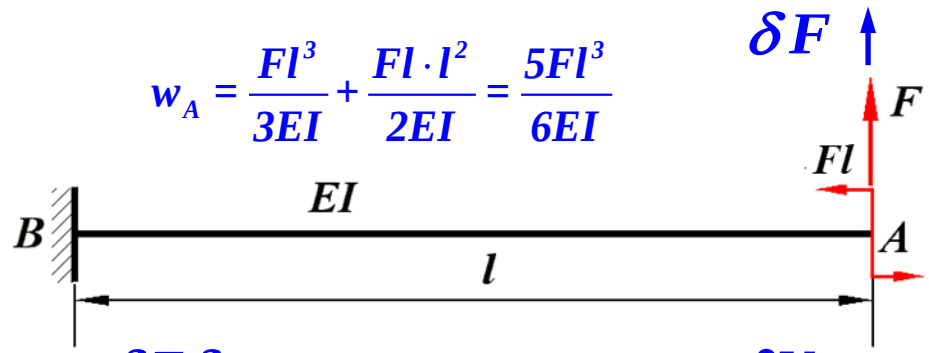
$$\frac{\partial M(x)}{\partial F}=x+l$$

力矩无增量，不能求偏导

$$w_A=\int_l \frac{M(x)}{EI}\frac{\partial M(x)}{\partial F}dx$$

$$=\int_0^l (x+l)^2\,\mathrm{d}x=\frac{7Fl^3}{3EI}$$?

$$w_A=\frac{Fl^3}{3EI}+\frac{Fl\cdot l^2}{2EI}=\frac{5Fl^3}{6EI}$$



$$\frac{\delta F\,\delta w_A}{2}+W_0+w_A\delta F=V_0+\frac{\partial V_\varepsilon}{\partial F}\delta F$$

区分载荷

$M_e=Fl$

$$M(x)=Fx+M_e \qquad \frac{\partial M(x)}{\partial F}=x$$

$$w_A=\frac{F}{EI}\int_0^l (x+l)\,x\,dx=\frac{F}{EI}\left(\frac{1}{3}l^3+\frac{l}{2}l^2\right)=\frac{5Fl^3}{6EI}(\downarrow)$$

理解相应位移

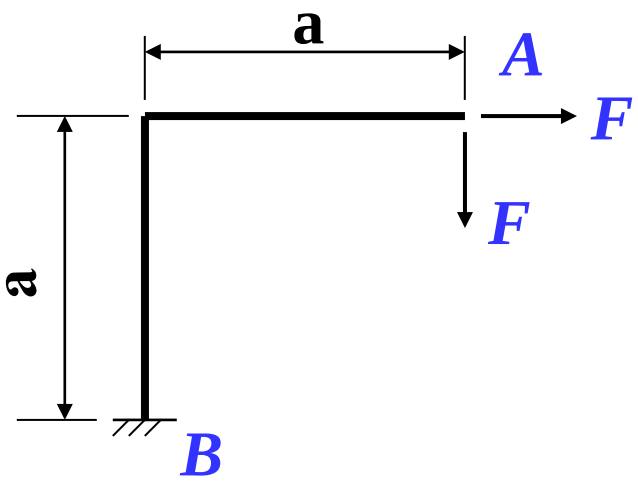
2、卡氏定理应用

3）应变能对广义载荷的偏导数

图示刚架EI为常数，分析$\dfrac{\partial V_\varepsilon}{\partial F}$的意义。

合力的相应位移？

A截面合位移？



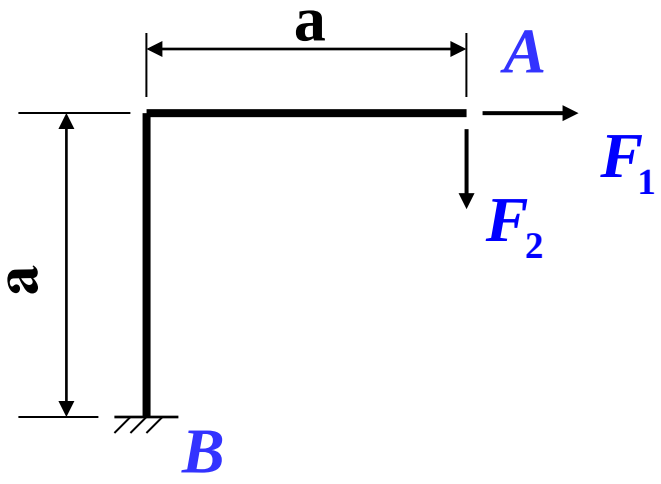
解：考虑该刚架的另一载荷状态，

其应变能

$$V_\varepsilon = V_\varepsilon(F_1, F_2)$$

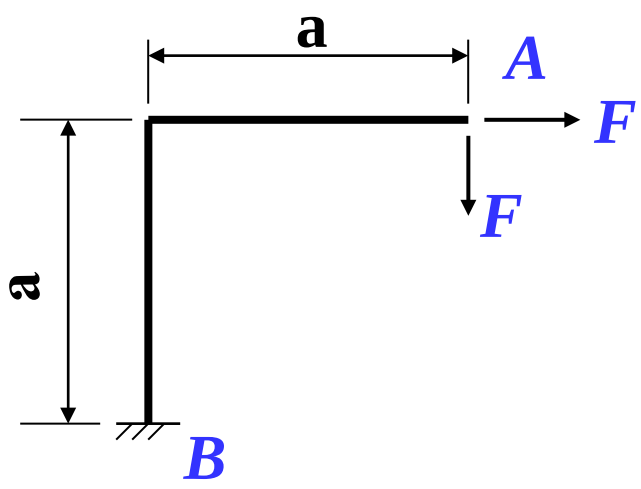
则

$$\frac{\partial V_\varepsilon}{\partial F_1} = \Delta_A, \qquad \frac{\partial V_\varepsilon}{\partial F_2} = f_A$$



2、卡氏定理应用

3）应变能对广义载荷的偏导数



若令 $F_1 = F$， $F_2 = F$

则由复合求导法则

$$\frac{\partial V_\varepsilon}{\partial F} = \frac{\partial V_\varepsilon}{\partial F_1}\frac{\partial F_1}{\partial F} + \frac{\partial V_\varepsilon}{\partial F_2}\frac{\partial F_2}{\partial F} = \frac{\partial V_\varepsilon}{\partial F_1} + \frac{\partial V_\varepsilon}{\partial F_2}$$

$$= \Delta_A\big|_{F_1=F} + f_A\big|_{F_2=F}$$

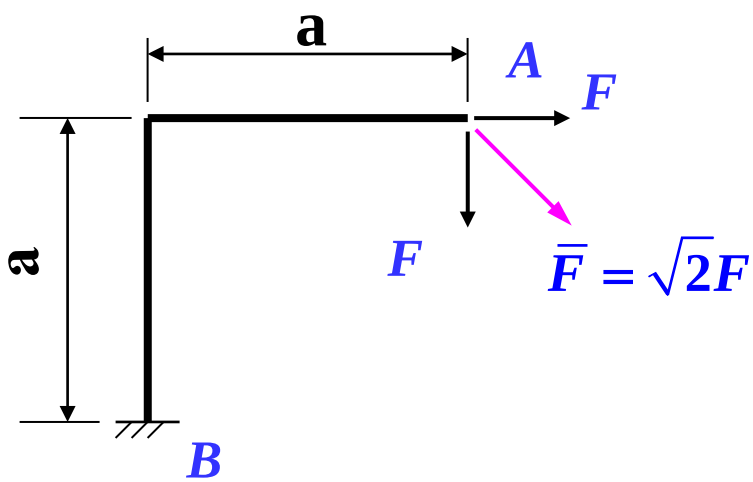
$\dfrac{\partial V_\varepsilon}{\partial F}$ 为两F载荷相应位移的代数和。

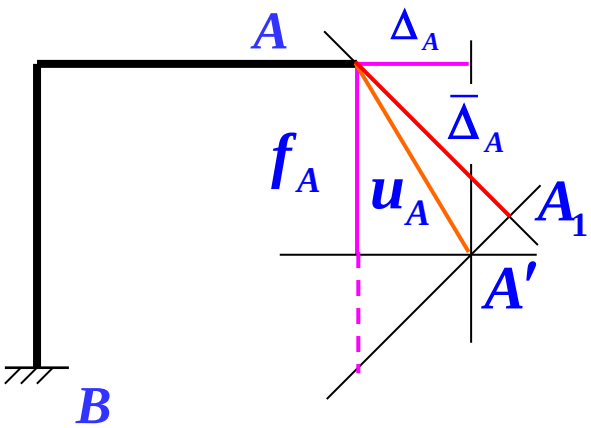
应用：若两个F载荷平行、反向，

$\dfrac{\partial V_\varepsilon}{\partial F}$ 为两载荷对应的相对位移。

2、卡氏定理应用

3）应变能对广义载荷的偏导数





合力的相应位移（A截面45度方向位移）

合位移

$$\bar{\Delta}_A=\frac{\sqrt{2}}{2}\left(\Delta_A+f_A\right)$$

$$u_A=\sqrt{\Delta_A^2+f_A^2}$$

$$\bar{\Delta}_A=\frac{\partial V_\varepsilon}{\partial \bar{F}}=\frac{\partial V_\varepsilon}{\partial\left(\sqrt{2}F\right)}=\frac{\sqrt{2}}{2}\frac{\partial V_\varepsilon}{\partial F}=\frac{\sqrt{2}}{2}\left(\Delta_A+f_A\right)$$

2、卡氏定理应用

小结：

- 附加载荷；
- 区分载荷；
- 应变能对广义载荷的偏导

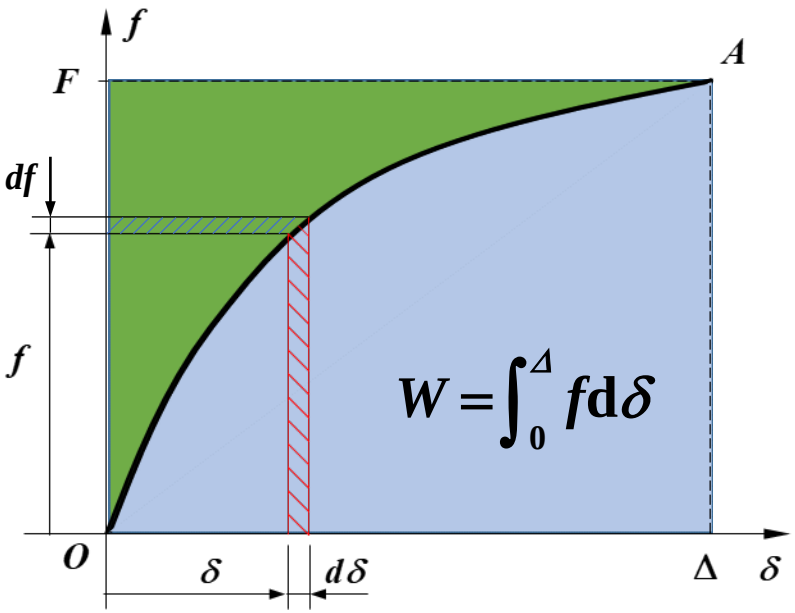
  ——求相对位移。

正确理解相应位移

3、余功与余能

余功的定义 Complementary Work

$$W_c = \int_0^F \delta \mathrm{d}f$$



余能（余应变能）

Complementary Energy

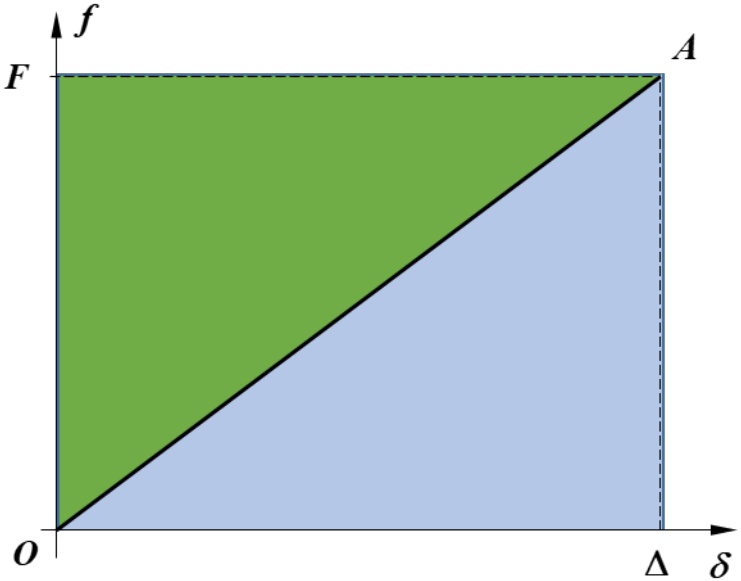
$$\boxed{V_c = W_c}$$

$$W_c + W = F\Delta$$

$$V_c + V_\varepsilon = F\Delta$$

线弹体

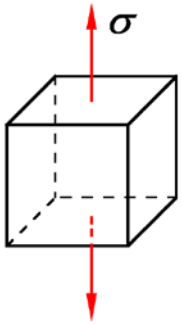
$$V_C = V_\varepsilon$$

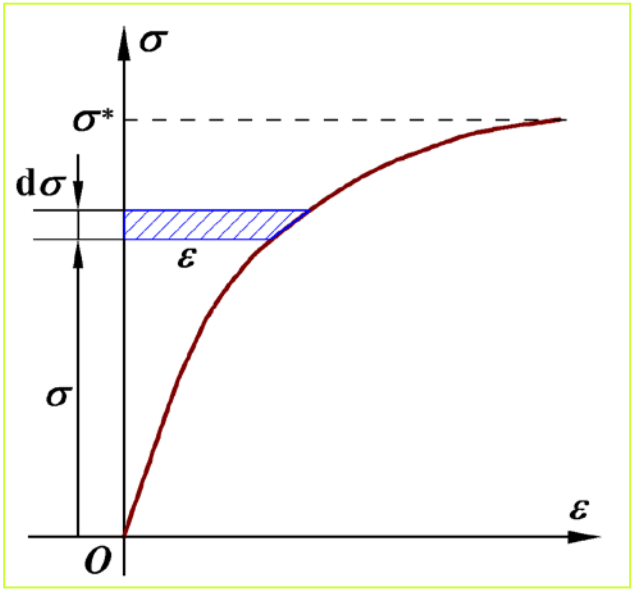


3、余功与余能

单向应力状态下的余能密度

$$v_c = \int_0^{\sigma^*} \varepsilon \mathrm{d}\sigma$$





拉压杆与梁的余能

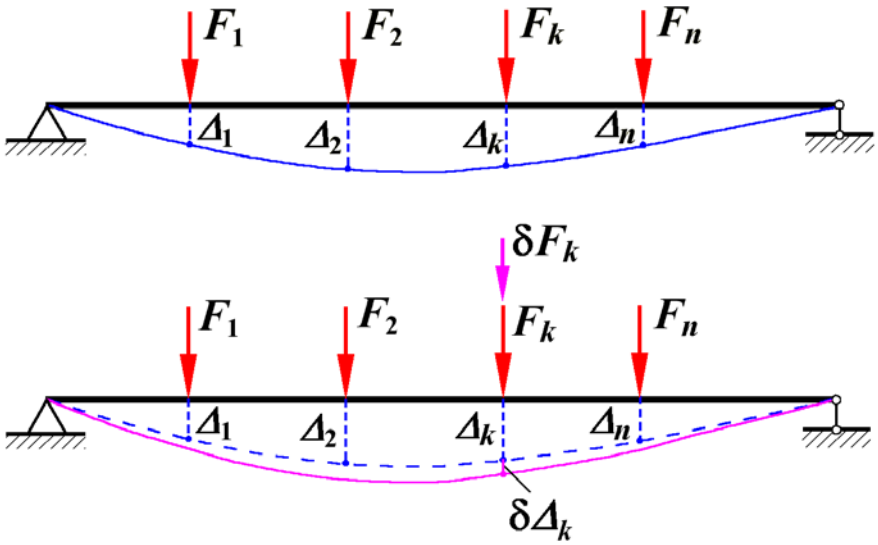
$$V_c = \int_V \int_0^{\sigma^*} \varepsilon \, \mathrm{d}\sigma \, \mathrm{d}V$$

余能是载荷的函数

$$V_C = V_C(F_1, F_2, \cdots, F_n)$$

4、克罗第-恩格塞定理

问题：弹性结构受n个载荷作用，求F_k的相应位移Δ_k



给载荷增量 δF_k

余功增量 $\delta W_c = \Delta_k \delta F_k$

余能增量 $\delta V_c = \dfrac{\partial V_c}{\partial F_k}\delta F_k$ $\quad V_C = V_C(F_1, F_2, \cdots, F_n)$

$$\Delta_k = \frac{\partial V_c}{\partial F_k}$$

弹性体的余能对载荷 F_k 的偏导数，等于该载荷的相应位移 Δ_k — 克罗第－恩格塞定理

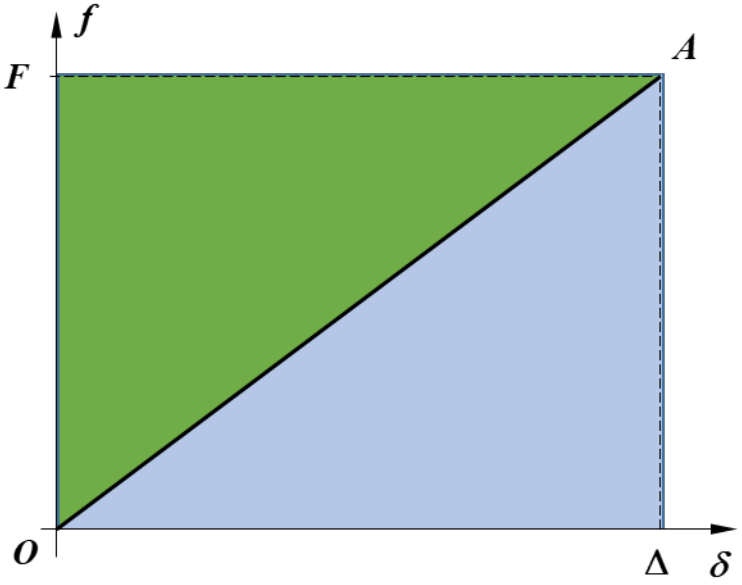
Crotti-Engesser's theorem

4、克罗第-恩格塞定理

对于线弹性体

$$V_C = V_\varepsilon$$

$$\boxed{\Delta_k = \frac{\partial V_\varepsilon}{\partial F_k}}$$

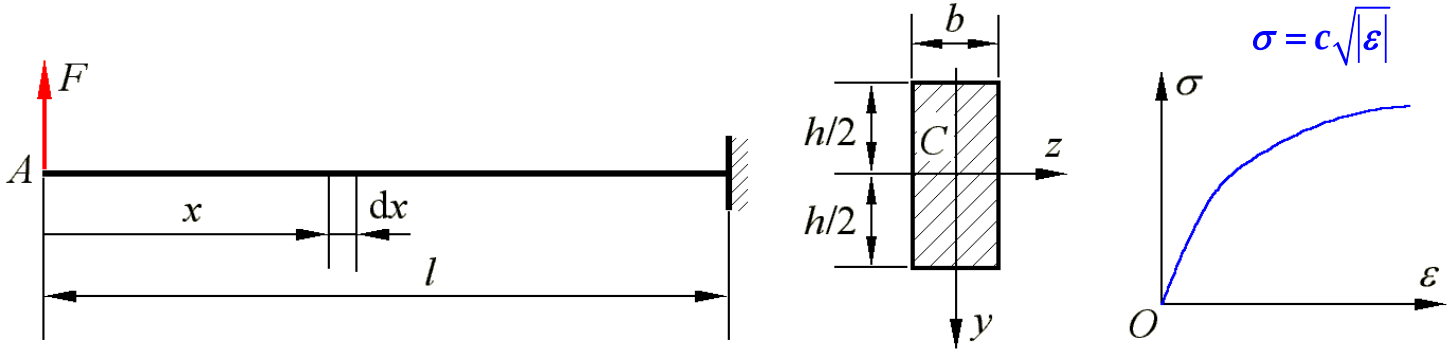


线性弹性体的应变能，对载荷 F_k 的偏导数，等于该载荷的相应位移 Δ_k

——卡氏第二定理

4、克罗第-恩格塞定理

例13-8 图示非线弹性悬臂梁，求自由端挠度w_A.



解：1）对称弯曲应力分析

平面假设与单向受力假设

$$\varepsilon = \frac{y}{\rho} \qquad \sigma = \begin{cases} \dfrac{c}{\sqrt{\rho}}\sqrt{y}, & y \geq 0; \\ -\dfrac{c}{\sqrt{\rho}}\sqrt{|y|}, & y < 0 \end{cases}$$

$$M = \int_A y\sigma \mathrm{d}A = 2\int_0^{h/2} \frac{c}{\sqrt{\rho}} y^{3/2} b\mathrm{d}y$$

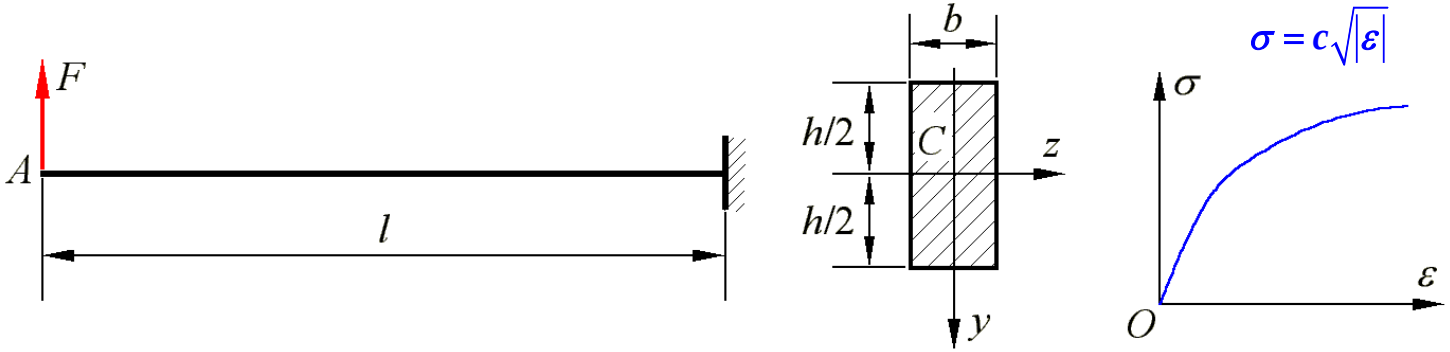
$$M = \frac{cbh^{5/2}}{5\sqrt{2\rho}} \qquad M(x) = Fx$$

$$\frac{1}{\sqrt{\rho}} = \frac{5\sqrt{2}Fx}{cbh^{5/2}}$$

$$\boxed{\sigma = \frac{5\sqrt{2}Fx}{bh^{5/2}}\sqrt{y}}$$

4、克罗第-恩格塞定理

例13-8 图示非线弹性悬臂梁，求自由端挠度w_A.



解：2）余能与位移计算

$$v_c = \int_0^{\sigma} \varepsilon \mathrm{d}\sigma = \int_0^{\sigma} \frac{\sigma^2}{c^2} \mathrm{d}\sigma$$

$$V_c = 2\int_0^{l}\int_0^{h/2} v_c \mathrm{d}y\mathrm{d}x = \frac{25F^3 l^4}{6c^2 b^2 h^5}$$

$$v_c = \frac{250\sqrt{2}F^3 y^{3/2} x^3}{3c^2 b^3 h^{15/2}}$$

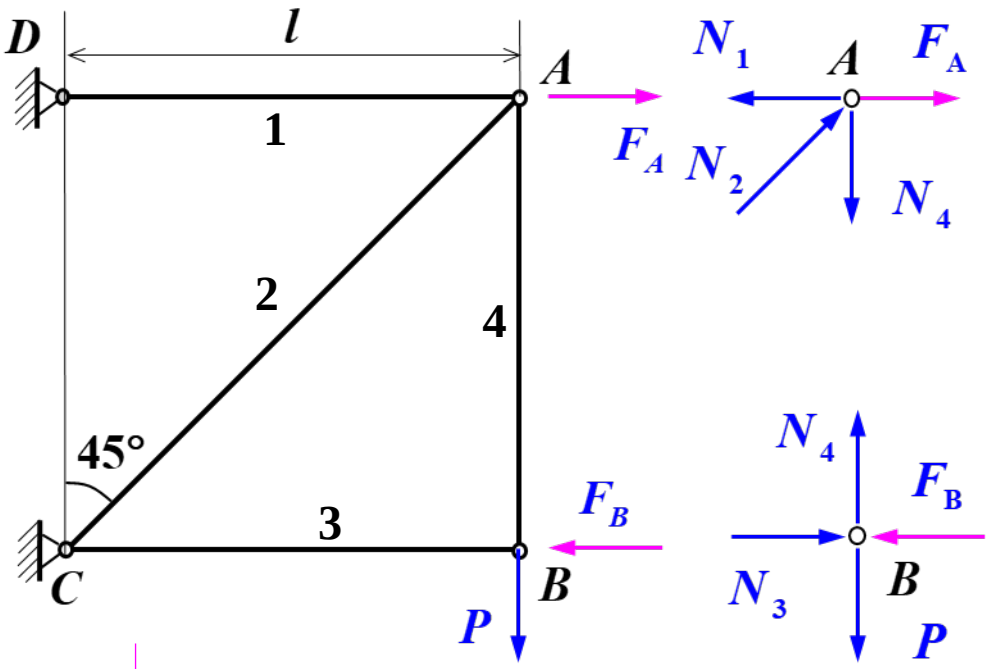
$$w_A = \frac{\partial V_c}{\partial F} = \frac{25F^2 l^4}{2c^2 b^2 h^5} \quad (\uparrow)$$

5、例题

例13-9 图示桁架，各杆EA相同，求：1）$\Delta_{A/B}$；2）θ_{AB}.

解I：分别求Δ_A和Δ_B

附加力F_A和F_B



$$N_3 = F_B, \quad N_4 = P$$

$$N_2 = \sqrt{2}P, \quad N_1 = F_A + P$$

$$\Delta_A = \frac{\partial V_\varepsilon}{\partial F_A} = \sum_{i=1}^{4} \frac{N_i l_i}{EA} \frac{\partial N_i}{\partial F_A}$$

$$\Delta_A = \frac{N_1 l_1}{EA} = \frac{(F_A + P)l}{EA}$$

$$\Delta_B = \frac{N_3 l_3}{EA} = \frac{F_B l}{EA}$$

节点水平位移等于水平杆件变形量

令， $F_A = 0, \quad F_B = 0$

$$\Delta_A = \frac{Pl}{EA}(\rightarrow), \quad \Delta_B = 0$$

$$\Delta_{A/B} = \Delta_A + \Delta_B, \quad \theta_{AB} = \frac{\Delta_{A/B}}{l}$$

5、例题

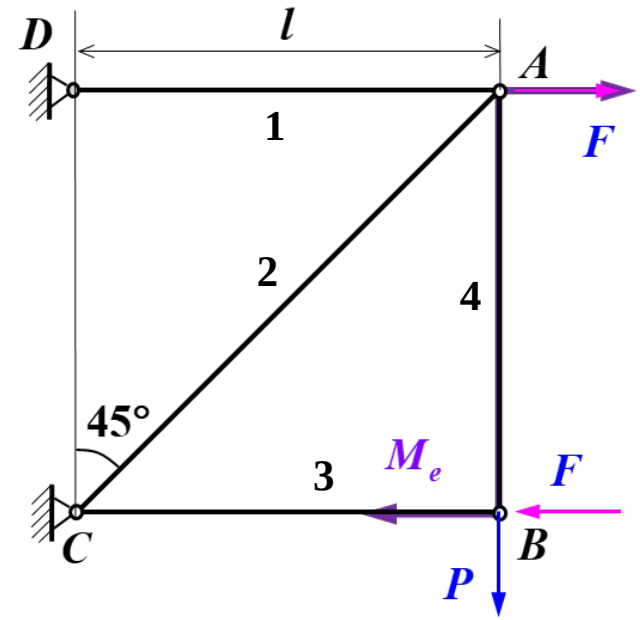
解II：直接求$\Delta_{A/B}$和θ_{AB}

附加一对力F

$$N_3 = F, \quad N_4 = P,$$

$$N_2 = \sqrt{2}P, \quad N_1 = F + P$$

$$\Delta_{A/B} = \frac{\partial V_\varepsilon}{\partial F} = \sum_{i=1}^{4} \frac{N_i l_i}{EA}\frac{\partial N_i}{\partial F} = \frac{N_1 l_1}{EA} + \frac{N_3 l_3}{EA}$$



附加力矩M_e=Fl

$$\theta_{AB} = \frac{\partial V_\varepsilon}{\partial M_e} = \frac{1}{l}\frac{\partial V_\varepsilon}{\partial F}$$

$$= \frac{\Delta_{A/B}}{l}$$

令， $F = 0$,

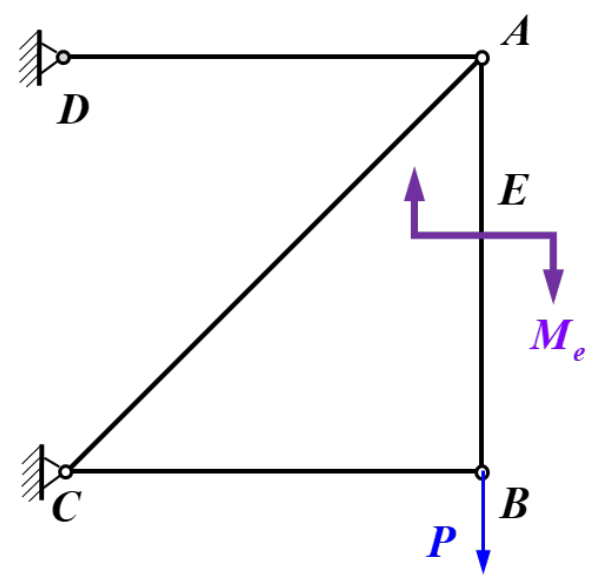
$$\Delta_{A/B} = \frac{Pl}{EA}, \quad \theta_{AB} = \frac{P}{EA}$$

思考：

任意处E加力矩M_e

1） θ_{AB}是M_e的相应位移吗？

2） $\theta_{AB} = \dfrac{\partial V_\varepsilon}{\partial M_e}$ ？

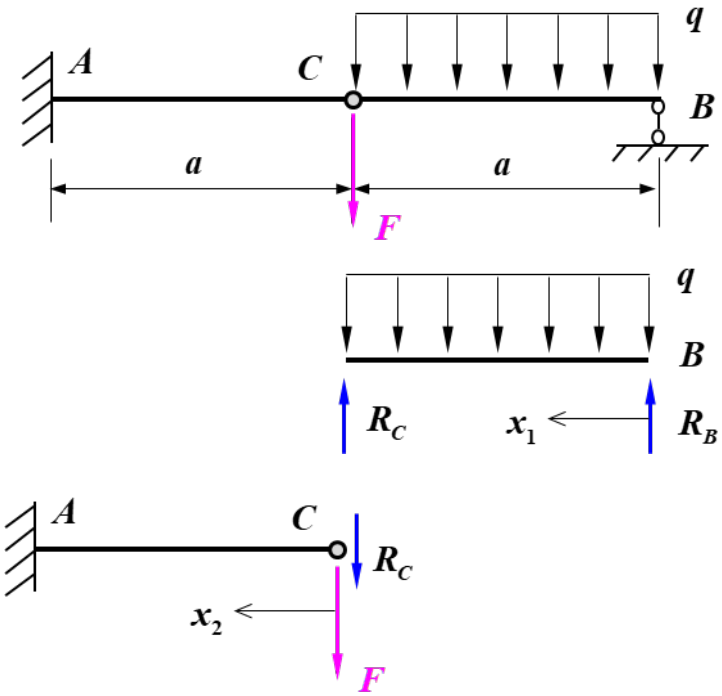


5、例题

例13-10 图示组合梁，EI为常数，求：f_C, $\theta_{CL/R}$.

解：

附加力F



$$R_B = R_C = \frac{qa}{2}$$

$$M_1(x_1) = R_B x_1 - q x_1 \frac{x_1}{2},$$

$$M_2(x_2) = -(F + R_C) x_2$$

$$\frac{\partial M_1}{\partial F} = 0, \quad \frac{\partial M_2}{\partial F} = -x_2$$

$$f_C = \int_0^a \frac{M_2}{EI} \frac{\partial M_2}{\partial F}\bigg|_{F=0} \mathrm{d}\, x_2 = \frac{R_C l^3}{3EI}$$

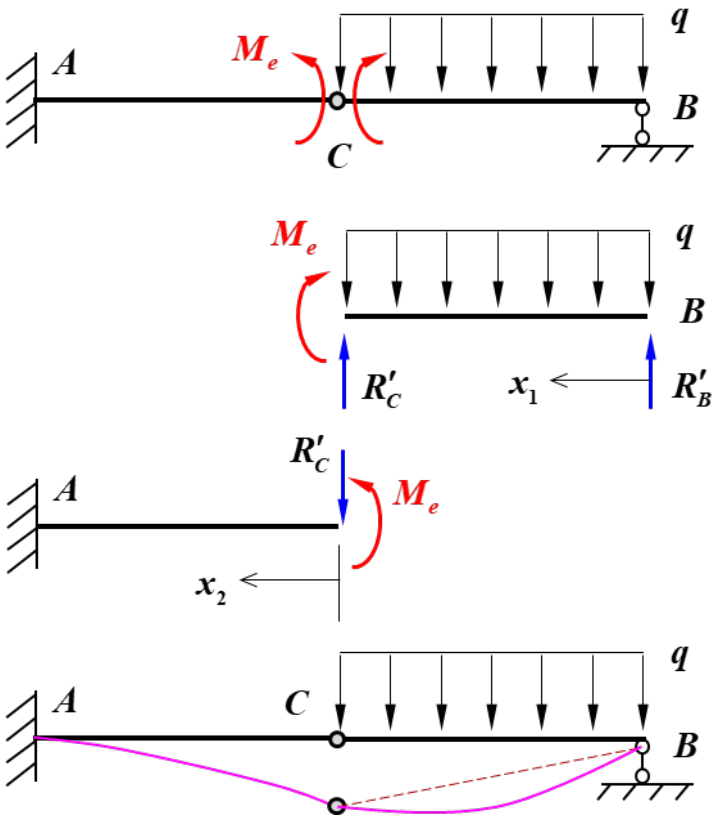
$$= \frac{qa^4}{6EI} (\downarrow)$$

C点挠度取决于AC梁弯曲

5、例题

例13-10 图示组合梁，EI为常数，求：f_C, $\theta_{CL/R}$.

解： 附加一对力矩M_e



$$R'_B = \frac{qa}{2} + \frac{M_e}{a}, \quad R'_C = \frac{qa}{2} - \frac{M_e}{a}$$

$$M_1(x_1) = R'_B x_1 - q x_1 \frac{x_1}{2}, \quad M_2(x_2) = M_e - R'_C x_2$$

$$\frac{\partial M_1}{\partial M_e} = \frac{x_1}{a}, \quad \frac{\partial M_2}{\partial M_e} = 1 + \frac{x_2}{a}$$

$$\theta_{CL/R} = \int_0^a \frac{M_1}{EI}\frac{x_1}{a}\mathrm{d}x_1 + \int_0^a \frac{M_2}{EI}\left(1 + \frac{x_2}{a}\right)\mathrm{d}x_2$$

$$= \frac{qa^3}{24EI} - \left(\frac{qa^3}{4EI} + \frac{qa^3}{6EI}\right)$$

$$= -\frac{3qa^3}{8EI}$$

对照叠加法，理解各积分项的含义

本讲：卡氏定理

附加载荷，区分载荷，求相对位移

下讲：虚功原理与单位载荷法